

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA0305	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
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Section / Batch:	B. Tech – AI	Date:	8/8/2026

Code of Conduct

I _____nagendra /192472399_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System

Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3
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NLP – MCQ

UNIT I – Introduction and Morphology

1. What is the primary goal of Natural Language Processing (NLP)?

- A) To increase computer hardware speed
- B) To enable computers to process and understand human language ✓**
- C) To design operating systems
- D) To manage computer networks

Answer: B) To enable computers to process and understand human language

2. Which model is commonly used to describe whether a string belongs to a particular language?

- A) Database model
- B) Mathematical model ✓**
- C) Financial model
- D) Hardware model

Answer: B) Mathematical model

3. Which symbol is commonly used in regular expressions to represent zero or more occurrences of the preceding element?

- A) +
- B) ?
- C) * ✓**
- D) ^

Answer: C) *

4. Which regular expression represents a sequence containing one or more occurrences of a?

- A) a*
- B) a+ ✓**
- C) a?
- D) a|

Answer: B) a+

5. What is the main function of a Finite State Automaton (FSA)?

- A) To store large databases
- B) To recognize patterns in strings ✓**
- C) To translate programming languages
- D) To generate images

Answer: B) To recognize patterns in strings

6. Which type of morphology deals with grammatical variations such as tense and number?

- A) Derivational morphology
- B) Inflectional morphology ✓**
- C) Computational morphology
- D) Structural morphology

Answer: B) Inflectional morphology

7. Which example represents derivational morphology?

- A) cat → cats
- B) walk → walked
- C) happy → happiness ✓
- D) book → books

Answer: C) happy → happiness

8. What is the purpose of finite-state morphological parsing?

- A) To identify the internal morphological structure of words ✓
- B) To identify network protocols
- C) To calculate sentence length
- D) To generate numerical data

Answer: A) To identify the internal morphological structure of words

9. What does the Porter Stemmer primarily perform?

- A) Sentence parsing
- B) Word stemming ✓
- C) POS tagging
- D) Speech recognition

Answer: B) Word stemming

10. The word "unhappiness" can be analyzed as un + happy + ness. Which process is illustrated by this analysis?

- A) Inflectional morphology only
- B) Derivational morphology ✓
- C) Syntax parsing
- D) Regular expression matching

Answer: B) Derivational morphology

UNIT II – Word Level and Syntactic Analysis

11. What does an N-gram model estimate?

- A) Probability of a word based on previous words ✓
- B) Number of letters in a word
- C) Meaning of a paragraph
- D) Number of sentences in a document

Answer: A) Probability of a word based on previous words

12. A bigram model considers how many words at a time?

- A) One
- B) Two ✓
- C) Three
- D) Four

Answer: B) Two

13. What is the main problem with an unsmoothed N-gram model?

- A) It cannot count words
- B) It assigns zero probability to unseen N-grams ✓
- C) It cannot recognize nouns
- D) It cannot process sentences

Answer: B) It assigns zero probability to unseen N-grams

14. What is the primary purpose of smoothing in language models?

- A) To remove punctuation
- B) To assign probabilities to unseen events ✓
- C) To identify verbs
- D) To shorten sentences

Answer: B) To assign probabilities to unseen events

15. What does entropy measure in an N-gram language model?

- A) Linguistic uncertainty ✓
- B) Number of nouns
- C) Sentence length
- D) Number of characters

Answer: A) Linguistic uncertainty

16. Which word class does "quickly" belong to in the sentence "She runs quickly"?

- A) Noun
- B) Verb
- C) Adjective
- D) Adverb ✓

Answer: D) Adverb

17. What is the purpose of a tagset in POS tagging?

- A) To define the set of grammatical labels used for words ✓
- B) To store sentences
- C) To generate regular expressions
- D) To calculate entropy

Answer: A) To define the set of grammatical labels used for words

18. In the sentence "Book a ticket," what is the POS of "Book"?

- A) Noun
- B) Verb ✓**
- C) Adjective
- D) Adverb

Answer: B) Verb

19. Which POS tagging approach uses manually written linguistic rules?

- A) Stochastic tagging
- B) Rule-based tagging ✓**
- C) Transformation-based tagging
- D) Random tagging

Answer: B) Rule-based tagging

20. Which POS tagging method combines an initial tagging with a sequence of corrective transformations?

- A) Rule-based tagging
- B) Stochastic tagging
- C) Transformation-based tagging ✓**
- D) Dictionary-based tagging

Answer: C) Transformation-based tagging

UNIT III – Context-Free Grammars

21. What is the primary purpose of a Context-Free Grammar (CFG) in NLP?

- A) To represent syntactic structure of sentences ✓
- B) To calculate word frequency
- C) To identify word meanings
- D) To translate documents

Answer: A) To represent syntactic structure of sentences

22. Which component of a CFG specifies how a non-terminal can be expanded?

- A) Terminal symbol
- B) Production rule ✓
- C) Probability table
- D) Lexicon only

Answer: B) Production rule

23. In the rule $S \rightarrow NP VP$, what does S usually represent?

- A) Sentence ✓
- B) Subject
- C) Semantics
- D) String

Answer: A) Sentence

24. Which structure visually represents the syntactic organization of a sentence?

- A) Bar chart
- B) Parse tree ✓
- C) Flowchart
- D) Frequency table

Answer: B) Parse tree

25. What does grammatical agreement ensure?

- A) Words have the same number of letters
- B) Related words have compatible grammatical features ✓
- C) All words have the same meaning
- D) Sentences contain equal numbers of words

Answer: B) Related words have compatible grammatical features

26. What does subcategorization describe?

- A) The arguments or complements that a word, especially a verb, requires ✓
- B) The number of letters in a word
- C) The probability of a sentence
- D) The meaning of punctuation

Answer: A) The arguments or complements that a word, especially a verb, requires

27. What is parsing in NLP?

- A) Converting text into numbers only
- B) Determining the syntactic structure of a sentence ✓
- C) Removing stop words
- D) Translating words between languages

Answer: B) Determining the syntactic structure of a sentence

28. Which parsing strategy begins with the start symbol and attempts to generate the input sentence?

- A) Bottom-up parsing
- B) Top-down parsing ✓**
- C) Stochastic tagging
- D) Semantic parsing

Answer: B) Top-down parsing

29. Which parsing algorithm can parse sentences using dynamic programming and handle left-recursive grammars?

- A) Porter algorithm
- B) Earley parser ✓**
- C) Viterbi tagger
- D) PageRank algorithm

Answer: B) Earley parser

30. What is the purpose of a Probabilistic Context-Free Grammar (PCFG)?

- A) To associate probabilities with grammar productions ✓**
- B) To remove grammar rules
- C) To identify word stems
- D) To classify documents

Answer: A) To associate probabilities with grammar productions

UNIT IV – Semantic Analysis

31. What is the main objective of semantic analysis?

- A) To determine the meaning of linguistic expressions ✓
- B) To count characters
- C) To identify punctuation
- D) To format documents

Answer: A) To determine the meaning of linguistic expressions

32. Which formal system is commonly used to represent relationships between entities and properties?

- A) First-Order Predicate Calculus ✓
- B) Regular Expression
- C) Finite Automaton
- D) N-gram model

Answer: A) First-Order Predicate Calculus

33. In first-order predicate logic, what does $P(x)$ typically represent?

- A) A probability
- B) A predicate applied to an argument ✓
- C) A parsing rule
- D) A POS tag

Answer: B) A predicate applied to an argument

34. Which representation can express the statement "John loves Mary" as a relationship between entities?

- A) loves(John, Mary) ✓
- B) John + Mary
- C) John/Mary
- D) John = Mary

Answer: A) loves(John, Mary)

35. What is syntax-driven semantic analysis?

- A) Deriving meaning representations based on syntactic structure ✓
- B) Counting syntactic words
- C) Removing grammatical rules
- D) Generating random meanings

Answer: A) Deriving meaning representations based on syntactic structure

36. What is a lexeme?

- A) An abstract vocabulary unit representing related word forms ✓
- B) A punctuation mark
- C) A sentence structure
- D) A grammar rule

Answer: A) An abstract vocabulary unit representing related word forms

37. What does word sense refer to?

- A) The grammatical category of a word
- B) A particular meaning of a word ✓
- C) The number of letters in a word
- D) The pronunciation of a word

Answer: B) A particular meaning of a word

38. What is the main goal of Word Sense Disambiguation (WSD)?

- A) To determine the intended meaning of an ambiguous word ✓
- B) To identify sentence boundaries
- C) To count words
- D) To remove prefixes

Answer: A) To determine the intended meaning of an ambiguous word

39. In "The fisherman sat on the bank," which meaning of "bank" is most appropriate?

- A) Financial institution
- B) River side ✓
- C) Storage facility
- D) Computer memory

Answer: B) River side

40. What is the primary goal of Information Retrieval (IR)?

- A) To retrieve relevant information from a collection of documents ✓
- B) To generate grammar rules
- C) To identify speech sounds
- D) To stem every word

Answer: A) To retrieve relevant information from a collection of documents

UNIT V – Language Generation and Discourse Analysis

41. What does discourse analysis primarily study?

- A) Language beyond individual sentences and its context ✓
- B) Individual characters only
- C) Computer hardware
- D) Word spelling only

Answer: A) Language beyond individual sentences and its context

42. What is reference resolution?

- A) Determining what an expression such as "he" or "it" refers to ✓
- B) Finding the number of words in a sentence
- C) Identifying punctuation marks
- D) Translating a sentence

Answer: A) Determining what an expression such as "he" or "it" refers to

43. In the sentence "Ravi bought a car. He likes it," what does "He" refer to?

- A) Car
- B) Ravi ✓
- C) Likes
- D) It

Answer: B) Ravi

44. What does text coherence refer to?

- A) The meaningful and logical relationship among parts of a text ✓
- B) The number of words in a paragraph
- C) The spelling of words
- D) The number of punctuation marks

Answer: A) The meaningful and logical relationship among parts of a text

45. What is a discourse act or dialogue act?

- A) The communicative function performed by an utterance ✓
- B) The number of words in dialogue
- C) A grammatical error
- D) A punctuation rule

Answer: A) The communicative function performed by an utterance

46. Which of the following is an example of a dialogue act?

- A) Request ✓
- B) Noun
- C) Prefix
- D) Stem

Answer: A) Request

47. What is the main purpose of a conversational agent?

- A) To interact with users through natural language ✓
- B) To perform only mathematical calculations
- C) To store images
- D) To compile source code

Answer: A) To interact with users through natural language

48. What is the purpose of surface realization in language generation?

A) To convert an abstract representation into natural-language text ✓

B) To identify word senses

C) To parse a sentence

D) To calculate N-gram entropy

Answer: A) To convert an abstract representation into natural-language text

49. Which machine translation approach uses an intermediate language-independent representation?

A) Direct translation

B) Interlingua ✓

C) Word-for-word translation

D) Rule deletion

Answer: B) Interlingua

50. What do statistical approaches to machine translation primarily learn from?

A) Parallel and bilingual language data ✓

B) Only punctuation marks

C) Computer hardware specifications

D) Regular expressions alone

Answer: A) Parallel and bilingual language data

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Underst anding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Applicati on of Concept s	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____